

# Stage 4 — Causal checks and the bootstrap correction

Where the Shortcut Sits: supervisor meeting 4 of 6

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**In one sentence.** The real-artifact crossover survives covariate matching and regression adjustment in all three cohorts where matching is feasible (A2: 3 of 3 supported); moving the same real artifact inside and outside the ROI of the same image produces a crossover too, but a neutral paste of the same shape reproduces most of it, so only a small part — significant for hair and thyroid calipers — is attributable to the artifact’s content. The bootstrap correction widened intervals by a median factor of 1.55 and changed 178 of 2454 significance verdicts.

## Notation

|                  |                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Y \in \{0, 1\}$ | diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).                                                                                                                                 |
| $A \in \{0, 1\}$ | artifact present (hair, sonographer caliper, capsule debris, chest tube).                                                                                                                                |
| ROI              | region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in <b>green</b> ; artifacts in <b>red</b> .                                                                                        |
| $r$              | overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $ . Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).                                                                           |
| Environments     | training and correlated test $P(A=1 \mid Y=1) = 0.9$ , $P(A=1 \mid Y=0) = 0.1$ ; reversed test 0.1/0.9; clean test: artifact independent of $Y$ (real traps: half of each class; sweeps: absent).        |
| ERM, mask        | linear head on the frozen features of the original image / of the ROI-masked image.                                                                                                                      |
| Crossover        | $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images). |
| CI               | 95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.                                                                                                   |
| $ \Delta p $     | same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.                                                                                                 |
| SMD              | standardised mean difference between two groups.                                                                                                                                                         |

## 1 Where we are

Stage 3 showed a positive crossover with real artifacts. Real artifacts are not placed at random: in dermoscopy, hair lies on large lesions more often than on small ones; in thyroid ultrasound, calipers inside the nodule are drawn on larger, more heavily measured nodules. Artifact-based debiasing studies in dermoscopy make the same point about where artifacts occur (Bissoto et al., 2020; Nauta et al., 2022). The crossover could therefore reflect *which images* carry in-ROI artifacts rather than *where* the artifact lies. This meeting reports the three checks designed to separate the two — matching, regression adjustment on the matched sample (A2, new in this round) and

transplantation of the same artifact with a paste-edge control — and, because it concerns what every interval means, the full consequences of the bootstrap correction.

## 2 Question

- Q4a** Is the crossover explained by measured differences between in-ROI and out-of-ROI artifact images?
- Q4b** Does it survive when the artifact is moved within the *same* image, and how much of that is due to the artifact rather than to pasting something into the image?
- Q4c** Which earlier claims change when the uncertainty is estimated correctly?

**Why three checks** Matching removes measured imbalance but depends on the chosen covariates; regression on the matched sample (a doubly robust combination) protects against residual imbalance and model misspecification of either step alone; transplantation removes image-level confounding altogether but introduces its own artifact — the paste. The neutral paste (same mask shape, same compositing, filled with the donor’s own clean tissue) measures that paste effect. Rejected alternatives: inverse-probability weighting (extreme weights with standardised differences above 2 in dermoscopy and capsule) and synthetic control images (they would test Stage 2 again, not real artifacts).

## 3 Methods

**Balance and matching (R1)** Pre-specified covariates per cohort (dermoscopy: lesion area, hair amount, age, sex, site, source, image size; thyroid: nodule area, caliper pixels, image size, grey level, nodule position; ovary and capsule analogous). Standardised mean differences (SMD) between Trap A and Trap B artifact images; greedy 1:1 nearest-neighbour matching on the logit of a propensity model within label, without replacement, calliper 0.2 SD (0.05 SD as a stricter sensitivity analysis, R7). A cohort with fewer than 30 matched images per label is infeasible.

**Regression adjustment (A2)** For each reversed-test image of the matched sample, the *placement value* of the masked minus the ERM score (for a positive: the weighted share of negatives ranked below it; for a negative: the complement of the share of positives ranked above it), so that the class means reproduce the AUROC and the unadjusted difference between traps equals the crossover exactly (verified per seed). Within each seed, weighted least squares of this outcome on an intercept, the Trap-B indicator, the label and all matching covariates; the adjusted crossover is the Trap-B coefficient averaged over seeds. Crossed bootstrap with placement values and regression recomputed in every replicate (2,000 replicates); Holm over the three cohorts.

**Transplant (R2) and neutral paste (R5)** For each artifact-free recipient image, a real artifact instance is cut from a donor of the same cohort (the donor’s mask pixels, largest connected component) and alpha-composited at a position fully inside and at a position fully outside the recipient’s ROI, with a random dihedral transform. The trap environments are then built as in Stage 1 with the pasted instance as the artifact. The neutral paste keeps the mask shape, the transform and the compositing but fills the mask with tissue from the donor image at a window that does not touch its artifact.

**Bootstrap correction (A1)** Every archived interval with a regenerated counterpart was compared row by row: old and new interval, whether “excludes zero” changed, width ratio and point-estimate difference.

## 4 Experiments

Table 1: Experiments of this stage. Commit times come from this machine and are not independent proof of order; the final registration was additionally pushed to GitHub before the runs.

| Experiment                                    | Status                           | Registration | First commit (local) | Support criterion |                                                        |
|-----------------------------------------------|----------------------------------|--------------|----------------------|-------------------|--------------------------------------------------------|
| Covariate balance (SMD) and matched traps, M1 | <b>supported</b> registered      | REVIEW2 (R1) | 29cd0da, 18:56       | 2026-09-27        | matched crossover CI > 0, Holm over 3 feasible cohorts |
| Stricter calliper 0.05 SD (R7)                | <b>supported</b> registered      | REVIEW3      | 85da735, 23:43       | 2026-09-27        | sensitivity, descriptive                               |
| Real-artifact transplant, T1                  | <b>supported</b> registered      | REVIEW2 (R2) | 29cd0da, 18:56       | 2026-09-27        | interaction CI > 0, Holm over 4                        |
| Neutral paste (paste-edge control), N3        | <b>supported</b> registered      | REVIEW3 (R5) | 85da735, 23:43       | 2026-09-27        | artifact – neutral interaction CI > 0                  |
| Regression-adjusted crossover, A2             | <b>supported</b> registered      | FINAL (A2)   | c45d482, 00:35       | 2026-09-28        | adjusted crossover CI > 0, Holm over 3                 |
| Crossed bootstrap everywhere, A1              | <b>supported</b> registered      | FINAL (A1)   | c45d482, 00:35       | 2026-09-28        | rewrite claims whose CI now includes 0                 |
| Capsule matched analysis                      | <b>not supported</b> descriptive | REVIEW2 (R1) | 29cd0da, 18:56       | 2026-09-27        | infeasible under the registered rule                   |

## 5 Results

### 5.1 Imbalance and matching

The in-ROI and out-of-ROI artifact images differ strongly (Table 2): in dermoscopy the largest SMD is 2.127 (lesion area: hair lies inside large lesions), in capsule 2.574 (roi area). Matching brings the maximum SMD to 0.108 (dermoscopy), 0.045 (thyroid) and 0.079 (ovary); capsule matching is infeasible (19 pairs in the smaller label, SMD after matching 0.231). The crossover is essentially unchanged by matching (Table 3): dermoscopy +0.154 [+0.123, +0.186], thyroid +0.238 [+0.186, +0.290], ovary +0.163 [+0.050, +0.273]; with the stricter calliper +0.124 [+0.086, +0.162], +0.218 [+0.165, +0.270] and +0.152 [+0.041, +0.265]. Capsule, matched descriptively, gives +0.230 [+0.036, +0.434] — lower than the unmatched +0.377 [+0.327, +0.426] and imprecise.

Table 2: Balance of pre-specified covariates between the in-ROI (Trap A) and out-of-ROI (Trap B) artifact images, before and after 1:1 nearest-neighbour matching within label (calliper 0.2 SD, pooled standardised mean differences). Registered rule: fewer than 30 matched images per label in either trap makes a cohort infeasible; target max |SMD| ≤ 0.1 after matching.

| Cohort    | Least balanced covariate | max  SMD  before | after matching | min matched pairs per label | feasible (rule) |
|-----------|--------------------------|------------------|----------------|-----------------------------|-----------------|
| ISIC hair | lesion_area              | 2.127            | 0.108          | 398                         | yes             |
| Thyroid   | caliper_px_log           | 0.527            | 0.045          | 93                          | yes             |
| Ovary     | caliper_px_log           | 0.389            | 0.079          | 86                          | yes             |
| Capsule   | roi_area                 | 2.574            | 0.231          | 19                          | <b>no</b>       |

Table 3: Crossover before and after covariate matching (reversed-test AUROC, crossed 95% CIs). The 0.05 SD calliper is the stricter sensitivity analysis of the third review (R7).

| Cohort    | Crossover, all images   | matched (0.2 SD)        | matched (0.05 SD)       | M1 (Holm, 3)             |
|-----------|-------------------------|-------------------------|-------------------------|--------------------------|
| ISIC hair | +0.147 [+0.111, +0.185] | +0.154 [+0.123, +0.186] | +0.124 [+0.086, +0.162] | <b>supported</b>         |
| Thyroid   | +0.239 [+0.196, +0.280] | +0.238 [+0.186, +0.290] | +0.218 [+0.165, +0.270] | <b>supported</b>         |
| Ovary     | +0.170 [+0.087, +0.252] | +0.163 [+0.050, +0.273] | +0.152 [+0.041, +0.265] | <b>supported</b>         |
| Capsule   | +0.377 [+0.327, +0.426] | +0.230 [+0.036, +0.434] | +0.328 [+0.087, +0.573] | descriptive (infeasible) |

Table 4: Per-covariate balance, thyroid and ovary (pooled over labels). The dermoscopy and capsule tables (17 and 20 covariates) are in `results/rerun_2026-09-28/review2/R1_balance.csv`.

| Cohort  | Covariate        | SMD before | SMD after  |
|---------|------------------|------------|------------|
| Thyroid | roi_area         | +0.395     | <i>n/a</i> |
| Thyroid | caliper_px_log   | +0.527     | <i>n/a</i> |
| Thyroid | width            | −0.143     | <i>n/a</i> |
| Thyroid | height           | −0.123     | <i>n/a</i> |
| Thyroid | mean_grey        | +0.183     | <i>n/a</i> |
| Thyroid | roi_centroid_row | +0.060     | <i>n/a</i> |
| Ovary   | roi_area         | +0.320     | <i>n/a</i> |
| Ovary   | caliper_px_log   | +0.389     | <i>n/a</i> |
| Ovary   | width            | −0.298     | <i>n/a</i> |
| Ovary   | height           | +0.047     | <i>n/a</i> |
| Ovary   | mean_grey        | +0.107     | <i>n/a</i> |
| Ovary   | subtype=0.0      | +0.149     | −0.079     |
| Ovary   | subtype=1.0      | +0.059     | +0.072     |
| Ovary   | subtype=2.0      | −0.159     | −0.012     |
| Ovary   | subtype=3.0      | −0.054     | +0.000     |
| Ovary   | subtype=4.0      | −0.022     | +0.024     |
| Ovary   | subtype=6.0      | +0.024     | +0.056     |
| Ovary   | subtype=7.0      | −0.020     | −0.062     |

## 5.2 Regression adjustment on the matched sample (A2)

The adjusted crossover equals the unadjusted one to the third decimal in every cohort (Table 5): dermoscopy +0.156 [+0.127, +0.187] (unadjusted +0.154 [+0.124, +0.186]), thyroid +0.238 [+0.184, +0.291] (+0.238 [+0.185, +0.291]), ovary +0.165 [+0.056, +0.273] (+0.163 [+0.058, +0.268]); Holm-adjusted  $p = 0.002$ ,  $0.002$  and  $0.002$ . All 3 pre-registered tests are supported. Residual imbalance after matching (up to the per-label maxima in Table 5) does not carry the effect. Capsule stays descriptive, as registered. What this cannot exclude is an *unmeasured* difference between the images.

Table 5: Regression-adjusted crossover on the matched traps (A2): per-image placement values of the masked minus the ERM score regressed on the Trap-B indicator, the label and all matching covariates, within seed; crossed bootstrap (2,000 replicates). Capsule is not analysed (matching infeasible; descriptive only).

| Cohort    | Test images per seed | Covariate columns | Post-matching max  SMD  pooled / per label | Unadjusted crossover    | Adjusted (doubly robust) | $p$ (Holm, 3) | A2               |
|-----------|----------------------|-------------------|--------------------------------------------|-------------------------|--------------------------|---------------|------------------|
| ISIC hair | 2960                 | 17                | 0.108 / 0.164                              | +0.154 [+0.124, +0.186] | +0.156 [+0.127, +0.187]  | 0.002         | <b>supported</b> |
| Thyroid   | 2150                 | 6                 | 0.045 / 0.150                              | +0.238 [+0.185, +0.291] | +0.238 [+0.184, +0.291]  | 0.002         | <b>supported</b> |
| Ovary     | 576                  | 11                | 0.079 / 0.140                              | +0.163 [+0.058, +0.268] | +0.165 [+0.056, +0.273]  | 0.002         | <b>supported</b> |

### 5.3 The same artifact inside and outside the ROI

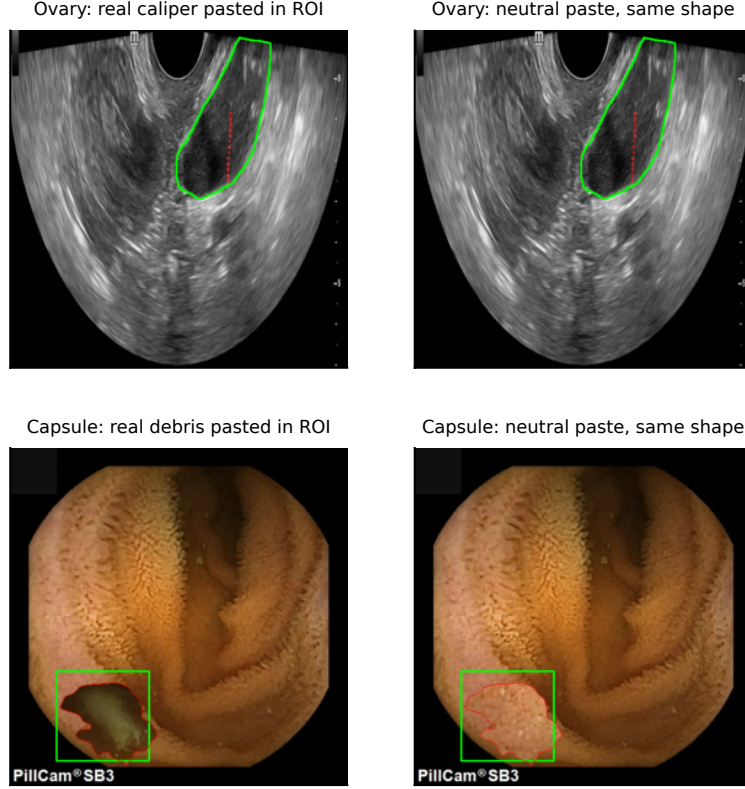


Figure 1: Transplant and neutral paste on the same recipient images (openly licensed cohorts; MMOTU, SEE-AI CC BY 4.0). Left: a real artifact instance pasted inside the ROI (green; the pasted region outlined in red). Right: the neutral paste — same mask shape and compositing, filled with artifact-free tissue from the donor.

Pasting the same real instance outside versus inside the ROI yields a positive location interaction in every cohort (Table 6; dermoscopy  $+0.253$  [ $+0.214, +0.293$ ], thyroid  $+0.277$  [ $+0.256, +0.298$ ], ovary  $+0.074$  [ $+0.044, +0.104$ ], capsule  $+0.497$  [ $+0.427, +0.567$ ]). The neutral paste, however, reproduces 52% to 92% of that interaction (Table 7): pasting *anything* into the ROI changes what masking does, because the paste itself is a new, label-correlated pattern inside the ROI (the environments make the pasted region label-correlated whatever it contains). Only the difference between the two — N3, the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content: dermoscopy  $+0.049$  [ $+0.026, +0.072$ ], thyroid  $+0.132$  [ $+0.116, +0.148$ ], ovary  $+0.022$  [ $-0.002, +0.048$ ] and capsule  $+0.037$  [ $-0.017, +0.093$ ] (2 of 4 excluding zero). **Consequently no harm figure from the transplant (for example, the effect of masking with the pasted hair inside the lesion) is attributed to the real artifact.** The transplant supports the *location* mechanism — a label-correlated pattern inside the ROI survives masking — and N3 shows that the real hair and caliper content adds to it.

Table 6: Real-artifact transplant (T1): the same real artifact instance, cut from a donor image, pasted outside and inside the ROI of the same artifact-free recipient images. These effects combine the artifact with the paste itself (seam, occluded tissue); Table 7 separates them.

| Cohort    | mask — ERM, pasted outside ROI | pasted inside ROI             | Location interaction          | Balanced: interaction         |
|-----------|--------------------------------|-------------------------------|-------------------------------|-------------------------------|
| ISIC hair | $+0.178$ [ $+0.128, +0.227$ ]  | $-0.075$ [ $-0.114, -0.036$ ] | $+0.253$ [ $+0.214, +0.293$ ] | $-0.052$ [ $-0.071, -0.033$ ] |
| Thyroid   | $+0.349$ [ $+0.321, +0.379$ ]  | $+0.072$ [ $+0.043, +0.100$ ] | $+0.277$ [ $+0.256, +0.298$ ] | $-0.016$ [ $-0.025, -0.007$ ] |
| Ovary     | $+0.064$ [ $+0.001, +0.126$ ]  | $-0.010$ [ $-0.075, +0.056$ ] | $+0.074$ [ $+0.044, +0.104$ ] | $-0.011$ [ $-0.025, +0.002$ ] |
| Capsule   | $+0.322$ [ $+0.262, +0.384$ ]  | $-0.175$ [ $-0.241, -0.109$ ] | $+0.497$ [ $+0.427, +0.567$ ] | $-0.032$ [ $-0.068, +0.004$ ] |

Table 7: Paste-edge control (R5): the same mask shape and compositing, filled with the donor’s own artifact-free tissue. Only the N3 column — the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content.

| Cohort    | mask – ERM, neutral outside | neutral inside          | Neutral interaction     | Artifact – neutral (N3) | Neutral / artifact interaction |
|-----------|-----------------------------|-------------------------|-------------------------|-------------------------|--------------------------------|
| ISIC hair | +0.129 [+0.079, +0.181]     | −0.075 [−0.116, −0.035] | +0.204 [+0.164, +0.244] | +0.049 [+0.026, +0.072] | 81%                            |
| Thyroid   | +0.200 [+0.175, +0.226]     | +0.055 [+0.028, +0.081] | +0.145 [+0.130, +0.161] | +0.132 [+0.116, +0.148] | 52%                            |
| Ovary     | +0.072 [+0.009, +0.135]     | +0.019 [−0.041, +0.080] | +0.052 [+0.021, +0.084] | +0.022 [−0.002, +0.048] | 71%                            |
| Capsule   | +0.215 [+0.154, +0.280]     | −0.244 [−0.314, −0.174] | +0.459 [+0.383, +0.535] | +0.037 [−0.017, +0.093] | 92%                            |

## 5.4 Consequences of the bootstrap correction

Across 2454 regenerated intervals with an archived counterpart, the corrected intervals are wider by a median factor of 1.55 (interquartile range 1.33–1.74; Figure 2); 146 lost significance and 32 gained it (Table 8). 160 regenerated point estimates differ from the archived ones by more than 0.03; most are in the capsule synthetic sweeps (up to 0.222), whose cohort was rebuilt on this machine: only 32 of the 71 archived artifact-bearing test frames fall in the new test set of 150 frames, because a seeded shuffle of the group list re-dealt the split. Table 9 lists the 33 verdict changes that touch a quoted claim. The claims rewritten as a result:

- Controlled sweeps: masking at 25% overlap (dermoscopy, DINOv2 at 518 and 224 px) and at 50% overlap (ovary) is no longer significantly harmful; MedSigLIP’s benefit at full thyroid overlap is no longer significant; the lesion-size strata lost two cells (Stage 2).
- Occlusion control: the small positive effects of masking on the 50/50 test are no longer significant; the conclusion (occlusion does not harm) is unchanged.
- Primary family: 11 instead of 12 of 16 supported after Holm (Stage 3); P3 ovary lost support.
- Natural test sets: masking’s gain on easy thyroid pairs and on MSK dermoscopy, and U-MtE’s gain over masking on hard thyroid pairs, are no longer significant (Stage 5).
- Fine-tuned models: several thyroid comparisons changed in both directions; fine-tuned results remain supplementary (Stage 5).
- Gained: masking’s cost in clean-test AUROC in the hair Trap A is now significant.

Table 8: Old versus corrected intervals by result group (every regenerated interval with an archived counterpart; `results/bootstrap_correction/sweeps_umte_old_vs_new.md`).

| Result group         | Intervals | Lost significance | Gained | $ \Delta $ estimate > 0.03 | max $ \Delta $ estimate | median width ratio |
|----------------------|-----------|-------------------|--------|----------------------------|-------------------------|--------------------|
| PRIMARY_CLAIMS.csv   | 16.0      | 2.0               | 0.0    | 1.0                        | 0.067                   | 1.36               |
| adhoc_bootstrap.json | 45.0      | 1.0               | 0.0    | 6.0                        | 0.076                   | 1.54               |
| analysis             | 4.0       | 0.0               | 0.0    | 0.0                        | 0.015                   | 1.63               |
| capsule              | 390.0     | 22.0              | 9.0    | 21.0                       | 0.057                   | 1.58               |
| finetune             | 42.0      | 4.0               | 3.0    | 9.0                        | 0.062                   | 1.21               |
| natural              | 75.0      | 4.0               | 0.0    | 0.0                        | 0.000                   | 1.74               |
| ovary                | 252.0     | 11.0              | 0.0    | 0.0                        | 0.013                   | 1.66               |
| review2              | 60.0      | 1.0               | 0.0    | 0.0                        | 0.000                   | 1.63               |
| review3              | 55.0      | 0.0               | 0.0    | 0.0                        | 0.000                   | 1.00               |
| spec_e13             | 301.0     | 19.0              | 9.0    | 5.0                        | 0.056                   | 1.55               |
| synthetic            | 786.0     | 73.0              | 7.0    | 102.0                      | 0.222                   | 1.47               |
| thyroid              | 428.0     | 9.0               | 4.0    | 16.0                       | 0.098                   | 1.62               |

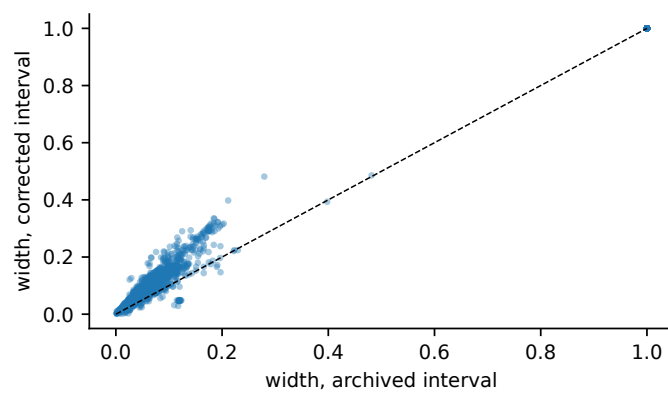


Figure 2: Width of each corrected interval against the archived one (dashed: equal width).

Table 9: Verdict changes among intervals that concern masking, the primary family, natural test sets, fine-tuning and the transplant. Only the corrected interval is shown; the archived interval is in the comparison file.

| Result                                                              | Row                                                                   | Corrected estimate [95% CI] | Significance |
|---------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------|--------------|
| PRIMARY_CLAIMS.csv                                                  | P2 U-MtE (protected) > mask   Capsule                                 | −0.006 [−0.018, +0.006]     | lost         |
| PRIMARY_CLAIMS.csv                                                  | P4 U-MtE+balanced > balanced                                          | +0.031 [−0.014, +0.077]     | lost         |
| adhoc_bootstraps.json                                               | Ovary<br>sweep   capsule_dino518   r=1.0   umte_<br>balanced-balanced | +0.057 [−0.008, +0.121]     | lost         |
| thyroid/convnext384_universal                                       | trapA   mask   clean   mask   erm                                     | +0.021 [−0.000, +0.041]     | lost         |
| synthetic/isc2018/dino518_ruler_<br>fixed_corr_main                 | 0.25   mask   erm   test_rev                                          | −0.059 [−0.127, +0.008]     | lost         |
| synthetic/isc2018/dino518_ruler_<br>fixed_occlusion_occlusion       | 0.0   mask   erm   test_uncorr                                        | +0.036 [−0.016, +0.086]     | lost         |
| synthetic/isc2018/dino518_ruler_<br>fixed_occlusion_occlusion       | 0.5   mask   erm   test_uncorr                                        | +0.041 [−0.010, +0.090]     | lost         |
| synthetic/isc2018/dino518_ruler_<br>fixed_occlusion_occlusion       | 0.75   mask   erm   test_uncorr                                       | +0.037 [−0.013, +0.086]     | lost         |
| synthetic/isc2018/dino518_ruler_<br>fixed_occlusion_occlusion       | 1.0   mask   erm   test_uncorr                                        | +0.039 [−0.010, +0.091]     | lost         |
| synthetic/isc2018/dino224_ruler_<br>fixed_corr_main                 | 0.25   mask   erm   test_rev                                          | −0.049 [−0.117, +0.018]     | lost         |
| synthetic/thyroid/medsiglip448_<br>caliper_corr_main                | 1.0   mask   erm   test_rev                                           | +0.044 [−0.011, +0.104]     | lost         |
| synthetic/ovary/dino518_caliper_<br>corr_main                       | 0.5   mask   erm   test_rev                                           | −0.107 [−0.269, +0.054]     | lost         |
| capsule/dino518_scale                                               | trapA   mask   clean   mask   erm                                     | +0.012 [−0.006, +0.031]     | lost         |
| spec_e13/dino518_matched05                                          | trapA   mask   clean   all   mask   erm                               | −0.026 [−0.053, +0.000]     | lost         |
| spec_e13/dino518_spec_scale                                         | trapB   mask   clean   all   mask   erm                               | +0.028 [−0.000, +0.055]     | lost         |
| spec_e13/dino518_spec_scale                                         | trapA   mask   test_rev   HAM   mask  <br>erm                         | −0.044 [−0.090, +0.002]     | lost         |
| spec_e13/dino518_spec_scale                                         | trapB   mask   clean   all   mask   erm                               | −0.025 [−0.055, +0.003]     | lost         |
| spec_e13/dino518_spec_universal                                     | trapA   mask   clean   all   mask   erm                               | −0.023 [−0.045, −0.001]     | gained       |
| spec_e13/dino518_spec                                               | trapA   mask   clean   all   mask   erm                               | −0.023 [−0.045, −0.001]     | gained       |
| spec_e13/dino518_spec_ablation                                      | trapA   mask   clean   all   mask   erm                               | −0.023 [−0.045, −0.001]     | gained       |
| spec_e13/dino518_spec_splice_<br>v2                                 | trapA   mask   clean   all   mask   erm                               | −0.023 [−0.045, −0.001]     | gained       |
| natural/thyroid_dino518_repro/<br>natural_boot.json                 | hard   mte-mask                                                       | +0.030 [+0.000, +0.061]     | lost         |
| natural/thyroid_dino518_repro/<br>natural_boot.json                 | easy   mask-erm                                                       | +0.042 [−0.011, +0.097]     | lost         |
| natural/isc_MSK_dino518_<br>repro/natural_boot.json                 | all   mask-erm                                                        | +0.016 [−0.007, +0.038]     | lost         |
| natural/isc_MSK_dino518_<br>repro/natural_boot.json                 | all   balanced-mask                                                   | −0.018 [−0.041, +0.005]     | lost         |
| finetune/thyroid/resnet50                                           | trapA   mte_post   mask_post                                          | −0.000 [−0.001, +0.001]     | lost         |
| finetune/thyroid/resnet50                                           | trapA   cons_ft   mask                                                | +0.042 [+0.011, +0.074]     | gained       |
| finetune/thyroid/resnet50                                           | trapB   mte_ft   mask                                                 | −0.042 [−0.089, +0.003]     | lost         |
| finetune/thyroid/resnet50                                           | trapB   mask_balanced   balanced                                      | +0.069 [−0.004, +0.143]     | lost         |
| finetune/thyroid/vit_small_<br>patch16_224.augreg_in21k_ft_<br>in1k | trapA   mask_balanced   balanced                                      | +0.059 [+0.003, +0.122]     | gained       |
| finetune/thyroid/vit_small_<br>patch16_224.augreg_in21k_ft_<br>in1k | trapA   umte_ft   mte_ft                                              | +0.032 [−0.072, +0.125]     | lost         |
| finetune/capsule/resnet50                                           | trapB   mte_ft   erm                                                  | +0.120 [+0.020, +0.217]     | gained       |
| review2/transplant/ovary_dino518                                    | 0.0   balanced   erm   test_rev                                       | +0.019 [−0.006, +0.046]     | lost         |

## 6 What failed or changed

- **The transplant “harm” is mostly a paste effect.** Earlier drafts quoted the pasted-hair effect as the harm of masking with in-lesion hair; that attribution is withdrawn. Only N3 is attributed to the artifact, and it is significant for hair and thyroid only.
- **Capsule cannot be matched** under the registered rule; its crossover is descriptive.
- **The bootstrap correction** changed 178 verdicts, and the claims above were rewritten.

- **The capsule synthetic cohort did not reproduce its split**; its point estimates moved by up to 0.222, while the location interaction kept its sign and significance.

## 7 Conclusion

**Supported.** The real-artifact crossover is not explained by measured covariates: it survives matching (M1, 3 of 3 feasible cohorts) and doubly robust adjustment (A2, 3 of 3, Holm). Moving the same real artifact inside the ROI changes what masking does, and for hair and thyroid calipers the artifact’s content adds to the effect of the paste. **Qualified.** Most of the transplant interaction is produced by the paste itself; unmeasured confounding of the natural location cannot be excluded. **Not supported.** A causal harm figure for the real artifact from the transplant; the capsule matched analysis.

## 8 Questions for the supervisor

1. Is matching plus A2 plus the transplant sufficient causal evidence for a Q1 journal, or should an instrumental or negative-control analysis be attempted?
2. How should the paper present the paste effect: as a limitation of the transplant, or as a finding in itself (any label-correlated pattern inside the ROI defeats masking)?
3. Should the full old-versus-new interval table be published as supplementary material?

## 9 Next stage

Stage 5 moves from trap environments to practice: natural test sets without an imposed correlation, clinical operating points, the thyroid subgroup of 78 patients, the breadth of replication across encoders and fine-tuning, the DermLIP exception, and a theory judged by its held-out prediction error.

## References

- Alceu Bissoto, Eduardo Valle, and Sandra Avila. Debiasing skin lesion datasets and models? not so fast. In *CVPR Workshops*, pages 740–741, 2020.
- Meike Nauta, Ricky Walsh, Adam Dubowski, and Christin Seifert. Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis. *Diagnostics*, 12(1):40, 2022.